# Stanley Nelavala

AI Software Engineer
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Huntington, IN | Open to relocate anywhere

## Summary

Software engineer who ships AI integrations end to end. Built the hospital design service with a deterministic layout engine, an on-device voice assistant with no cloud call, and a public LangGraph agent platform with human approval, 62 replay evals and failover between two LLM adapters. M.S. in Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate.

## Experience

**Software Development Engineer** Nov 2025 – Present
*Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test)* *Huntington, IN*

- Built RxDesign, the hospital design service, backend and web (FastAPI, React, TypeScript), with a deterministic, rule-based layout engine that places rooms on a 0.5 ft grid with doors, exits and flow lines; 220+ offline tests on the API.
- Built the RxAsset iOS and Android app (React Native, Expo) with scan-to-search and an on-device voice assistant (speech-to-text and text-to-speech run on the phone, no cloud call); fixed a native crash in the speech library bridge.
- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Replaced offset paging with keyset paging on the catalog device list: on a 328,000-device dev database, the query for row 150,000 fell from 285 ms to 0.20 ms. Search pages never skip or repeat rows, tested with rows inserted, deleted and tied between pages.
- Set up separate iOS and Android release pipelines in GitHub Actions with automatic TestFlight submission, and raised mobile statement coverage from 14% to 93% (582 tests).

**Adjunct Faculty (part-time)** Jan 2026 – Aug 2026
*The College of St. Scholastica* *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

**Graduate Research Assistant and Teaching Assistant** Aug 2023 – Aug 2025
*University of Minnesota Duluth* *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session. Led labs and office hours for Python, NLP and Parallel Programming.

## Projects (public code, tests)

**Agent Integration Platform** | *Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis, Terraform* GitHub

- LangGraph case workflow with typed state, a Postgres checkpointer, human approval that resumes after restart, and step and token budgets; one constrained MCP write tool.
- 62 deterministic replay evals, prompt-injection tests, output guardrails, PII redaction and an append-only audit log; full gate of 304 tests at 94% coverage.
- Hybrid retrieval (pgvector plus full-text search fused by RRF) with access labels; failover between two LLM adapters with timeouts, a retry budget, a circuit breaker and token cost records.

**Inference Gateway** | *Go, gRPC, ONNX Runtime, PyTorch, OpenTelemetry, Prometheus, Helm* GitHub

- Go service for an int8-quantized banking-intent classifier over HTTP and gRPC with micro-batching, deadlines and load shedding; macro-F1 0.8636 for both fp32 and int8 on the BANKING77 test split.
- One-factor experiments and pprof ruled out GC and tokenizer cost; a Little's law model sizes a 10x launch peak at 2 to 3 replicas (local benchmark).

## Skills

**AI and agents**: LangGraph, MCP, hybrid retrieval and RAG (pgvector, RRF), deterministic LLM evals, guardrails, PII redaction, LLM failover
**ML**: PyTorch, ONNX Runtime, int8 quantization, BERTopic, PaddleOCR, on-device speech (Whisper, Kokoro)
**Languages**: Python, TypeScript, Go, SQL, Java, Rust
**Backend and cloud**: FastAPI, PostgreSQL, Redis, gRPC, AWS (ECS, RDS, S3, SQS), Docker, Terraform, GitHub Actions
**Testing and observability**: pytest, Playwright, Locust, k6, OpenTelemetry, Jaeger, Prometheus

## Education and Certification

**M.S. Computer Science**, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
**B.Tech Computer Science and Engineering**, IIIT D&M Kancheepuram, 2019 – 2023
**AWS Certified Solutions Architect, Associate**, Feb 2025 – Feb 2028 (verifiable on Credly)